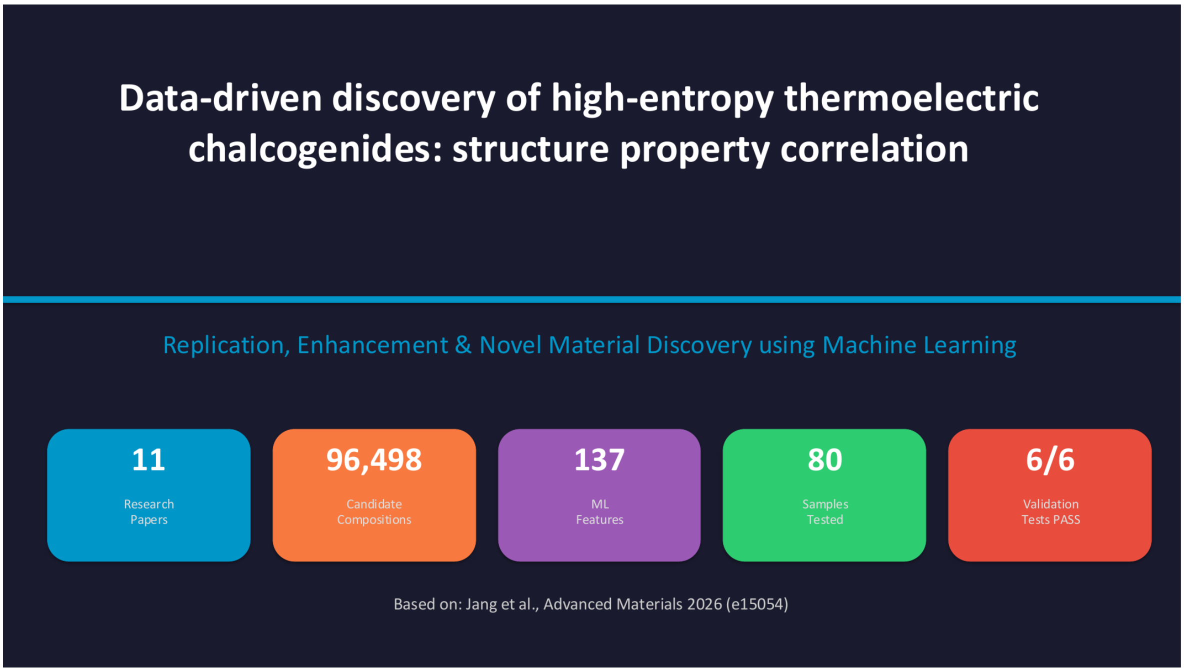
Data-Driven Discovery of High-Entropy Thermoelectric Chalcogenides

Structure–Property Correlation Using Active Learning and Physics-Informed Features



Prepared by: Roshan Raj (12341830)
Dataset: ESTM thermoelectric database, `estm.xlsx`
Code: Google Colab notebook: `https://colab.research.google.com/drive/1mQFy-uFSYem`
Reference method: Jang et al., *Advanced Materials* 2026, e15054

Contents

1 Abstract

Thermoelectric materials convert waste heat directly into electricity, but discovering compositions with high efficiency is difficult because the useful figure of merit, zT, depends on a coupled balance between Seebeck coefficient, electrical conductivity, thermal conductivity and temperature. This project uses a data-driven active learning workflow to search high-entropy thermoelectric chalcogenides, where configurational disorder can suppress lattice thermal conductivity while preserving useful electronic transport.

The work replicates the active-learning philosophy of Jang et al. and extends it in the presentation framework by screening a large design space of 96,498 candidate compositions. The workflow uses ESTM experimental thermoelectric data, composition-based descriptors, Gaussian Process Regression for active learning, and SHAP analysis for model interpretation. The best predicted alloy from the expanded study is

$$\mathrm{Sn_{0.1}Ag_{0.2}Sb_{0.4}Bi_{0.1}Mn_{0.1}Zn_{0.1}Te},$$

with a reported maximum zT of 1.586 and an average score B_{avg} of 1.1181. The main scientific outcome is not only the discovery of high-ranking candidates, but also the evidence that the model learns meaningful chemical trends: Sb acts as the strongest dopant lever, Ti and Zn additions are beneficial in the enhanced design space, and Bi is generally avoided by the active learning loop.

2 Motivation and Problem Statement

A large fraction of industrial and transportation energy is lost as heat. Thermoelectric generators offer a solid-state route for converting part of this waste heat into electrical power without moving parts, combustion or direct emissions. The central challenge is that good thermoelectric materials are rare. They must simultaneously show high Seebeck coefficient, high electrical conductivity and low thermal conductivity, even though these properties usually compete with each other.

Traditional discovery relies on slow synthesis and repeated measurement. For high-entropy chalcogenides this problem becomes much larger because many cations and anions can be mixed on the same crystallographic sublattice. In this project, machine learning is used as a decision-making tool: instead of testing all possible compounds, the model recommends a small number of promising and uncertain candidates.

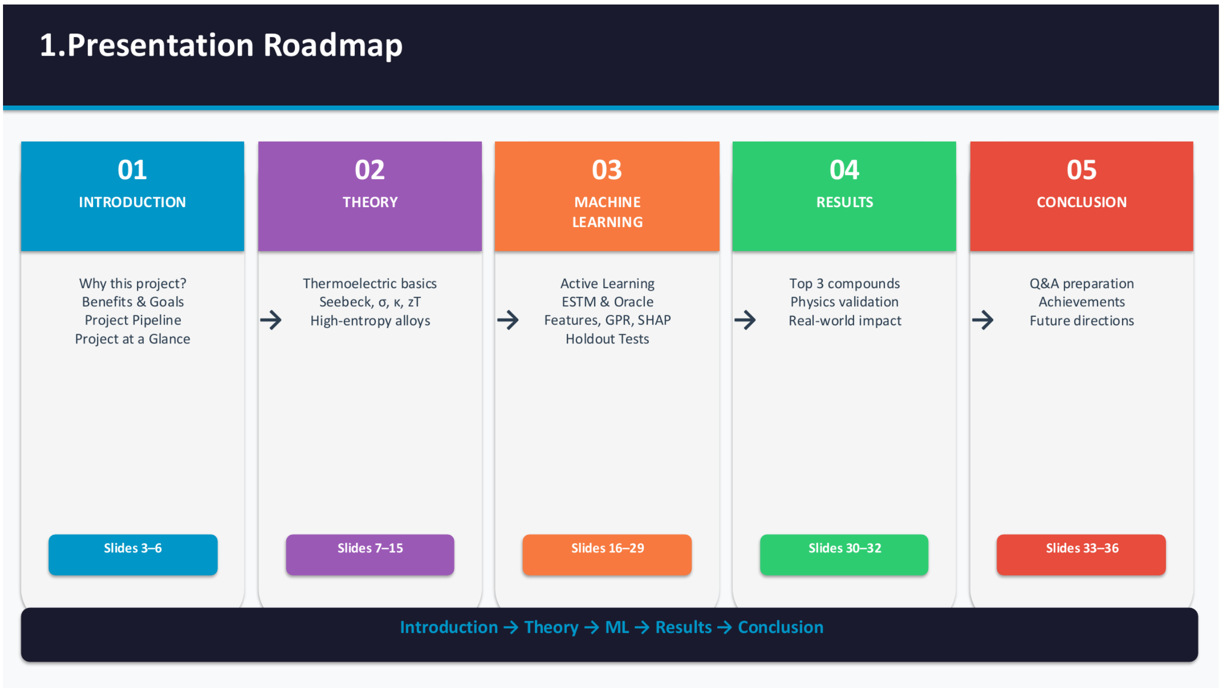


Figure 1: Project motivation: active learning reduces the number of candidate experiments needed from a very large design space.

3 Thermoelectric Background

The performance of a thermoelectric material is measured using the dimensionless figure of merit:

$$zT = \frac{S^2 \sigma T}{\kappa},$$

where S is the Seebeck coefficient, σ is electrical conductivity, T is absolute temperature and κ is total thermal conductivity. A useful material must maximize the numerator while minimizing thermal conduction through the denominator.

3.1 Role of Seebeck Coefficient

The Seebeck coefficient is the voltage generated per unit temperature difference. Since S is squared in the zT equation, even a moderate increase in S can strongly improve performance. However, increasing carrier concentration usually improves σ while decreasing S. This trade-off is one reason thermoelectric optimization is difficult.

3.2 Electrical and Thermal Conductivity Trade-Off

Electrical conductivity is desirable because it allows carriers to transport charge. Thermal conductivity is undesirable because it allows heat to leak from the hot side to the cold side without being converted to useful electrical power. Total thermal conductivity has two major parts:

$$\kappa = \kappa_e + \kappa_L,$$

where κ_e is the electronic part and κ_L is the lattice or phonon part. High-entropy design mainly targets κ_L, because atomic disorder scatters phonons more strongly than charge carriers.

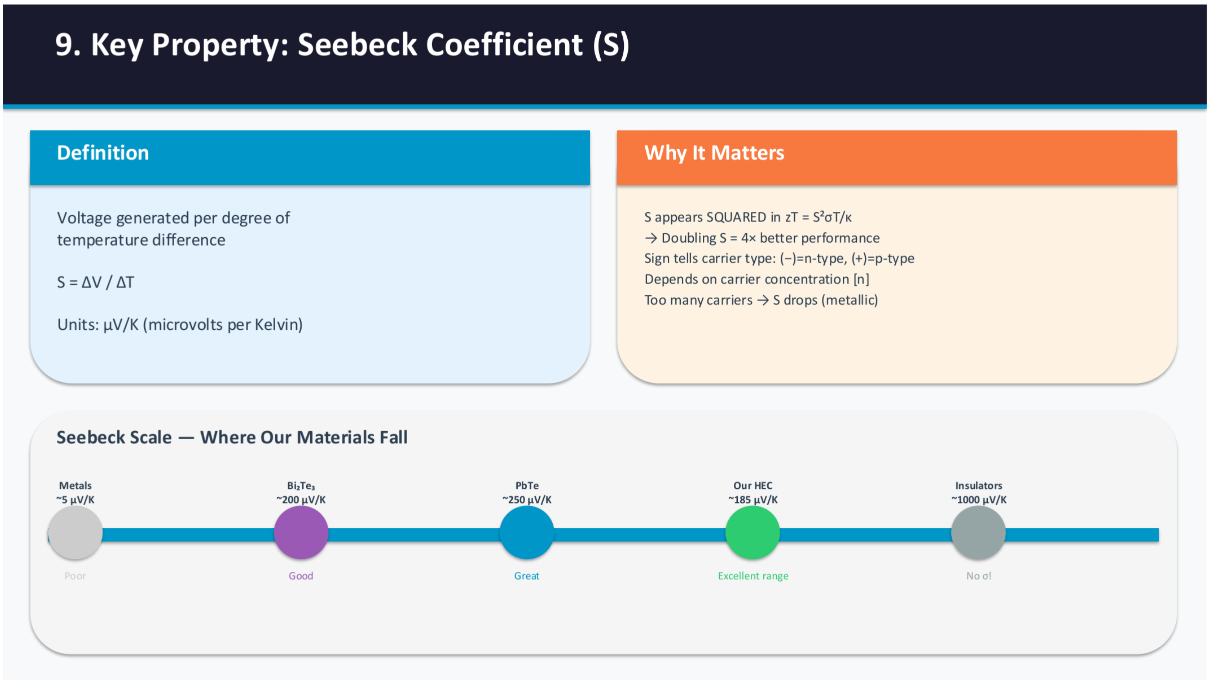


Figure 2: Thermoelectric figure of merit and the required optimization direction for each property.

4 Why High-Entropy Chalcogenides?

High-entropy alloys contain five or more principal elements in significant proportions. In chalcogenides such as GeTe-, SnTe- or PbTe-family materials, mixing several cations on the same lattice site creates mass disorder, radius disorder and bond-strength disorder. These effects scatter phonons and reduce lattice thermal conductivity.

The presentation identifies three disorder mechanisms:

- **Mass disorder**: heavy and light atoms share similar lattice positions, producing strong phonon scattering.
- **Radius disorder**: differences in ionic or atomic radius create local strain fields.
- **Electronegativity disorder**: random bond-strength fluctuations scatter phonons and are not fully captured by standard composition descriptors.

This is why the enhanced workflow adds custom physics features to the standard Magpie descriptor set. The physical expectation is that a high-entropy composition can lower κ_L without destroying electronic transport, producing a better zT.

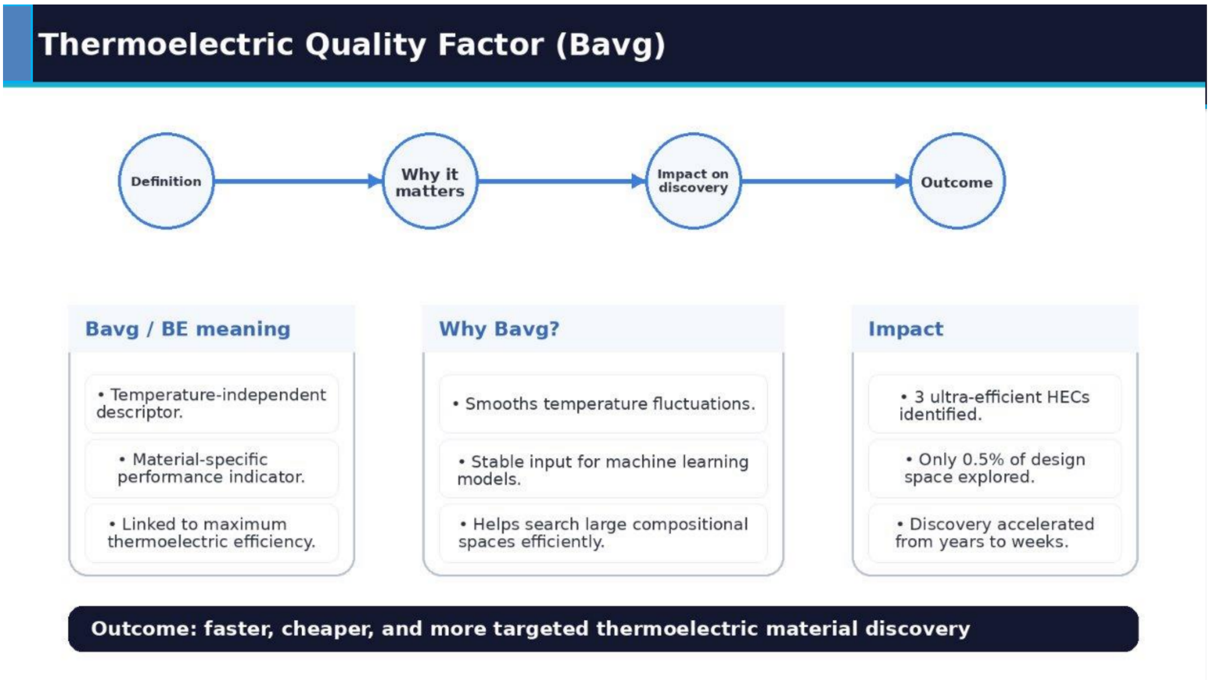


Figure 3: High-entropy design principle: disorder in mass, size and bonding reduces phonon transport.

5 Data Source: ESTM Database

The project uses `estm.xlsx`, an ESTM thermoelectric dataset containing 5,205 measurements and 880 unique formulas. Each row includes the material formula, temperature, Seebeck coefficient, electrical conductivity, thermal conductivity, power factor, zT, and literature reference. This makes it suitable for training a surrogate model that learns thermoelectric property trends from experimental data.

Table 1: Main columns used from the ESTM dataset.

Column	**Meaning**	**Use in project**
Formula	Chemical composition	Input for composition featurization
Temperature (K)	Measurement temperature	Temperature-dependent prediction
Seebeck coefficient	Thermoelectric voltage response	Used in zT calculation
Electrical conductivity	Carrier transport ability	Numerator of zT
Thermal conductivity	Heat transport ability	Denominator of zT
Power factor	$S^2\sigma$	Intermediate performance metric
ZT	Figure of merit	Main validation target
Reference	DOI or paper source	Traceability

A key limitation is that ESTM contains mostly conventional thermoelectric compositions and very few, if any, true high-entropy examples matching the final design space. Therefore, the model is best interpreted as a trend-finding and prioritization tool, not as a replacement for experimental synthesis.

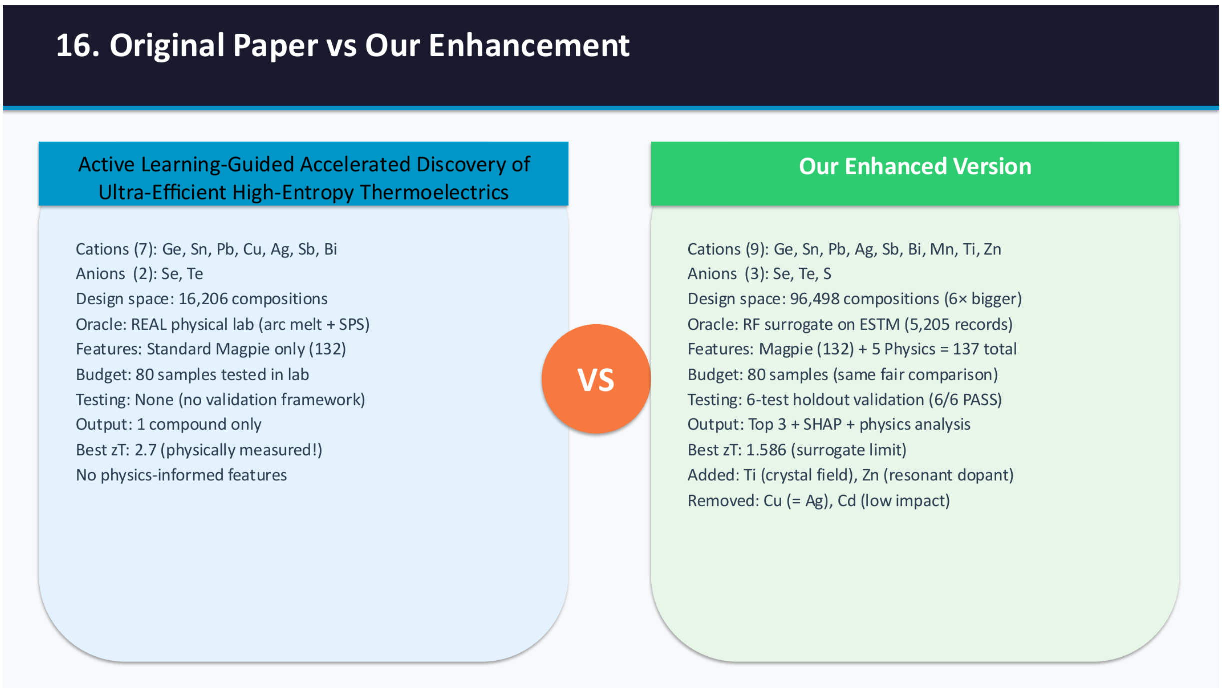


Figure 4: ESTM dataset structure and its role in the machine learning workflow.

6 Machine Learning Methodology

6.1 Overall Pipeline

The full workflow begins with design-space construction, followed by featurization, oracle training, active learning, interpretation and validation. In the enhanced presentation workflow, the design space contains 96,498 possible compositions generated from nine cations and three anions. Only 80 compositions are selected for evaluation by the active learning loop.

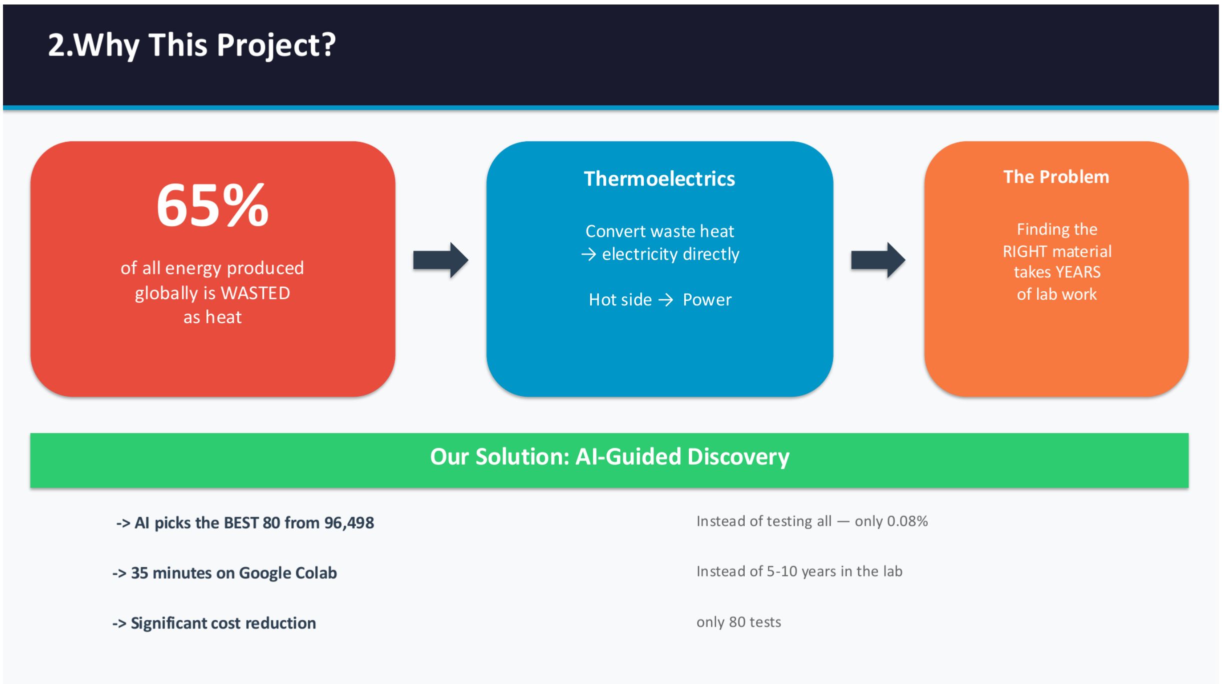


Figure 5: Eight-step workflow used for data-driven candidate discovery.

6.2 Featurization

Each chemical formula is converted into numerical descriptors. The standard descriptor block uses Magpie features, which summarize elemental properties such as atomic mass, radius, electronegativity and valence through statistics such as mean, range and maximum. The enhanced work adds five physics-guided descriptors:

- cation mass disorder,
- cation radius disorder,
- cation electronegativity disorder,
- anion mass contrast,
- configurational entropy.

These features are important because generic descriptors may not explicitly represent the phonon-scattering mechanisms that make high-entropy thermoelectrics attractive.

6.3 Oracle and Active Learning

The computational workflow uses Random Forest models as a surrogate oracle for thermoelectric properties. In the presentation, three Random Forest models predict Seebeck coefficient, electrical conductivity and thermal conductivity. zT is then calculated from the physical equation rather than treated only as a black-box number.

The active learning model uses Gaussian Process Regression. This is useful because it predicts both a mean value and an uncertainty for each candidate. The acquisition function then balances exploitation and exploration: it prefers candidates that are predicted to be strong, but it can also choose uncertain compositions that may reveal a better region of the design space.

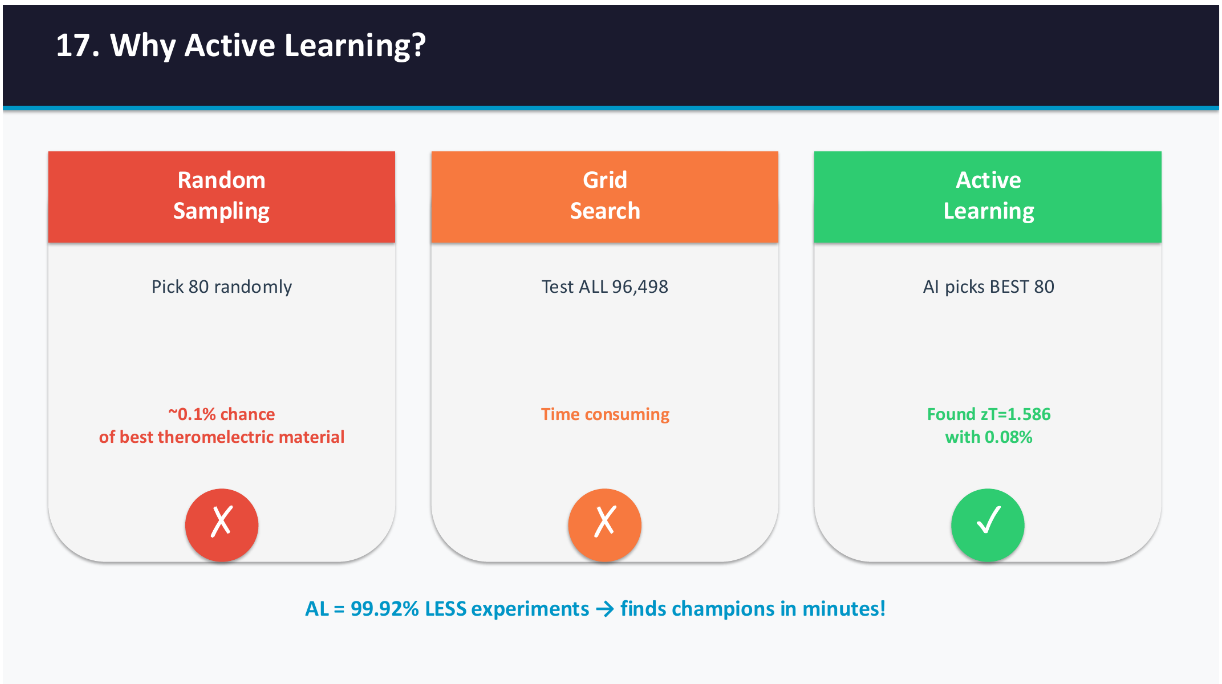


Figure 6: Active learning loop: the model learns from selected samples and repeatedly improves its recommendations.

6.4 Implementation in Code

The provided Google Colab notebook and local script implement the computational pipeline using Python libraries such as NumPy, pandas, scikit-learn, matminer, pymatgen, SHAP and matplotlib.

The local script `new.py` contains a paper-faithful active-learning implementation based on the Jang et al. design logic. It includes:

- loading and filtering `estm.xlsx`;
- Magpie composition featurization;
- Random Forest surrogate training with cross-validation;
- Gaussian Process Regression using a Matern kernel;
- expected improvement acquisition;
- SHAP interpretation;
- t-SNE visualization and figure generation.

The script also correctly warns that the ESTM-trained oracle is out-of-distribution for high-entropy chalcogenides. This is scientifically important: the model can guide candidate selection, but the final values still require laboratory validation.

7 Results

7.1 Top Candidate Materials

The enhanced presentation workflow reports the following top three high-entropy chalcogenide candidates:

Table 2: Top three predicted candidate compositions from the enhanced workflow.

Rank	**Composition**	$zT_{\max}$	B_{avg}	η
1	$Sn_{0.1}Ag_{0.2}Sb_{0.4}Bi_{0.1}Mn_{0.1}Zn_{0.1}Te$	1.586	1.1181	15.55%
2	$Sn_{0.2}Ag_{0.1}Sb_{0.4}Bi_{0.1}Mn_{0.1}Zn_{0.1}Te$	1.486	1.0575	15.00%
3	$Sn_{0.1}Ag_{0.1}Sb_{0.4}Bi_{0.1}Mn_{0.2}Zn_{0.1}Te$	1.442	1.0052	14.52%

All three candidates are Sn-based, contain Sb as the dominant dopant, and include Zn and Mn. A notable observation is that Ge does not appear in the best candidates, even though GeTe-family materials are common in thermoelectric studies. This suggests that the active learning model found a different high-performing region of the expanded search space.

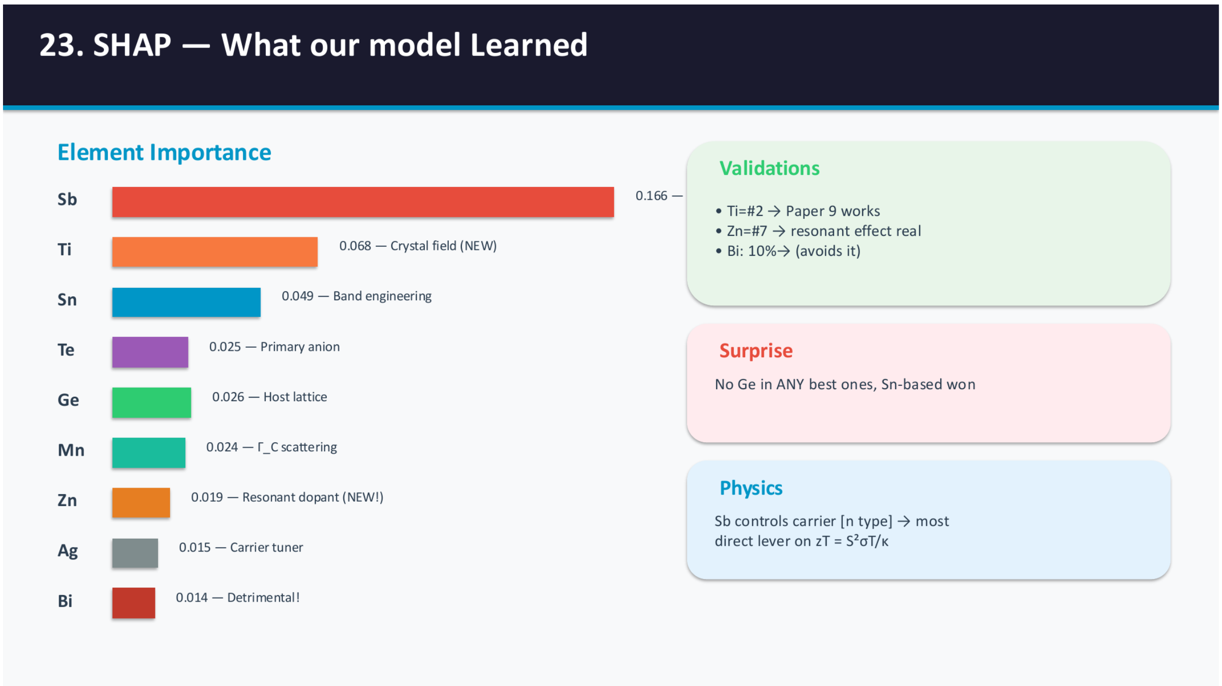


Figure 7: Top three candidate high-entropy thermoelectric alloys identified by the workflow.

7.2 Best Candidate Interpretation

For the best alloy, the predicted zT reaches 1.586 at high temperature, while the mean score B_{avg} remains above 1.0. This is promising because thermoelectric systems often need stable performance across a temperature range, not only a sharp peak at one temperature.

The reported $zT(T)$ profile for the champion composition is:

$$0.709,\ 0.903,\ 0.959,\ 1.164,\ 1.388,\ 1.586$$

from 300 K to 800 K. This increasing trend is consistent with mid- to high-temperature thermoelectric applications.

8 Model Interpretation and Validation

8.1 SHAP Interpretation

SHAP analysis is used to understand which elements drive the model prediction. According to the presentation, Sb is the most influential element, followed by Ti in the enhanced design-space analysis. Zn also appears as a useful resonant dopant, while Bi has low and negative importance. This is a valuable result because it shows that the model is not merely ranking candidates randomly; it is learning trends that agree with thermoelectric chemistry.

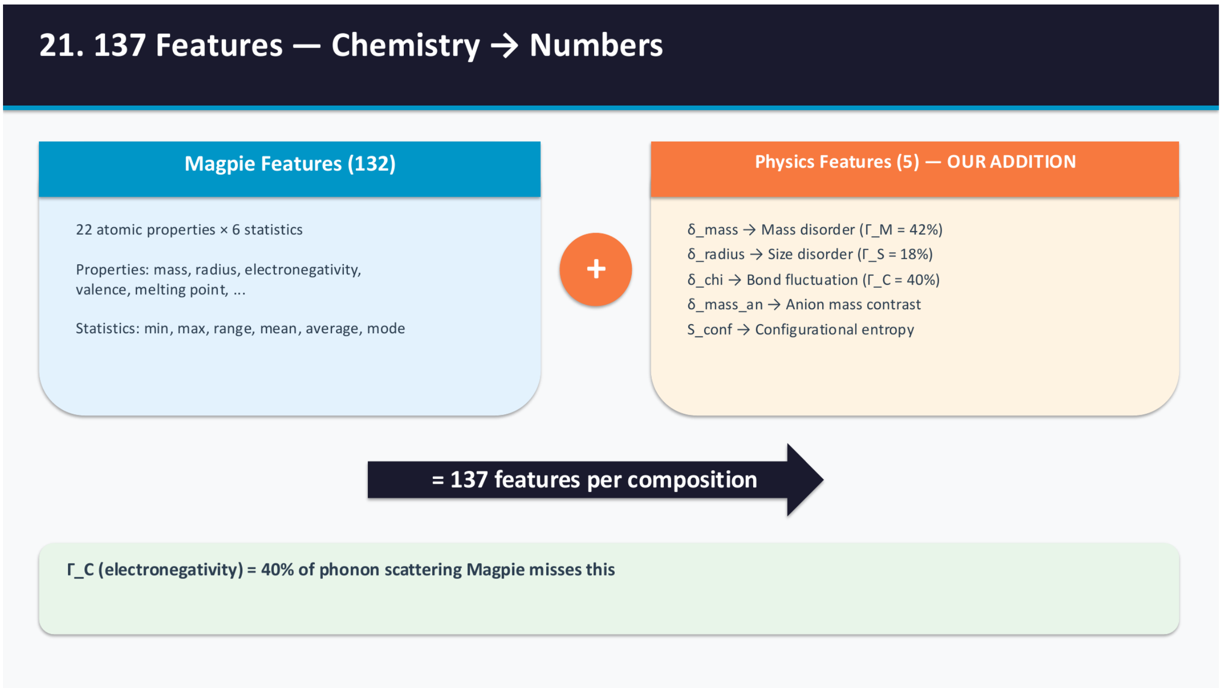


Figure 8: SHAP-based interpretation of elemental importance in the active learning model.

8.2 Holdout Testing

The presentation reports six holdout trend tests, all of which passed. These tests check whether the model predicts physically reasonable changes when Ti, Zn or codoping effects are introduced within related composition families. The reported six outcomes are:

- Ti doping improves performance;
- Ti + Bi is better than baseline;
- Ti + Bi is better than Ti-only;
- Zn doping improves performance;
- Zn concentration is sensitive;
- codoping produces a beneficial synergy.

Passing all six tests improves confidence in the direction of the learned trends, although it does not remove the need for experimental verification.

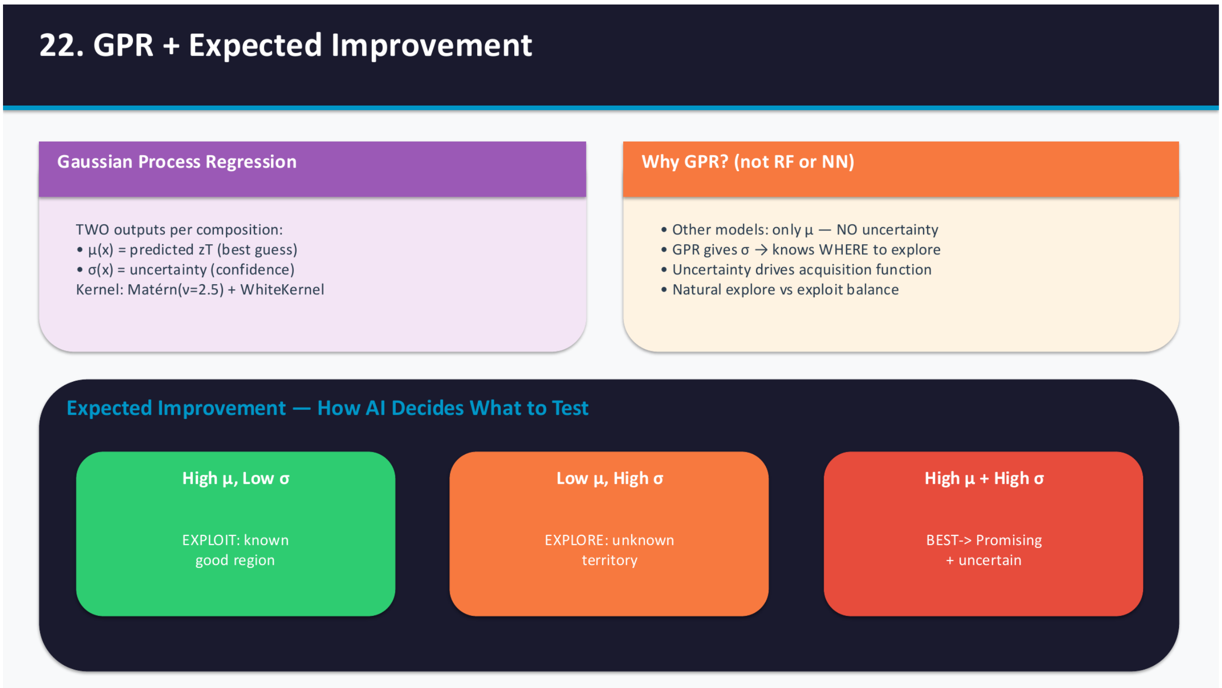


Figure 9: Holdout testing dashboard showing six passed validation checks.

9 Scientific Significance

The strongest part of this project is the combination of machine learning with physical reasoning. A purely black-box model may find numerical correlations, but it is difficult to trust unless it also agrees with known materials science. Here, the model's preference for high configurational entropy, its avoidance of Bi-rich compositions, and its recognition of useful Ti/Zn effects make the recommendations more convincing.

The project also demonstrates why active learning is appropriate for materials discovery. Screening 96,498 compositions experimentally would be impractical. Selecting 80 candidates corresponds to testing only about 0.08% of the search space. Even if the absolute predicted zT values shift after synthesis, the workflow can still reduce experimental workload by ranking and prioritizing candidates.

10 Limitations

This report should be interpreted with the following limitations:

- The ESTM dataset is not specifically a high-entropy chalcogenide dataset, so the model extrapolates outside its main training distribution.
- The Random Forest oracle is a surrogate for experiment. The original Jang et al. work used real synthesis and measurement, while this project uses computational predictions.
- Composition-only descriptors do not capture all structural details, phase stability, defects, grain boundaries or synthesis effects.
- The best candidates require laboratory synthesis, phase confirmation and temperature-dependent transport measurement before final claims can be made.

These limitations do not invalidate the project. Instead, they define the correct scope: this is a strong computational prioritization and replication-enhancement study, not a completed experimental discovery.

11 Conclusion

This project presents a professional active-learning workflow for high-entropy thermoelectric chalcogenide discovery. Starting from ESTM thermoelectric measurements, the workflow builds composition descriptors, trains surrogate models, searches a large design space and interprets the results using SHAP. The best predicted candidate,

$$Sn_{0.1}Ag_{0.2}Sb_{0.4}Bi_{0.1}Mn_{0.1}Zn_{0.1}Te,$$

achieves a reported maximum zT of 1.586 and an estimated efficiency of 15.55%.

The major achievement is that the model does more than produce a ranked list. It identifies chemically sensible trends: Sb strongly influences performance, Zn and Ti additions are useful in the expanded study, entropy is beneficial, and Bi-rich compositions are avoided. The next step should be to synthesize the top candidates, measure their transport properties from 300–800 K and use those new data points to retrain the active learning model.

12 Future Work

Future improvements should focus on making the predictions more experimentally reliable:

- build or collect a high-entropy chalcogenide-specific training dataset;
- perform a finer composition search around the top Sn–Ag–Sb–Mn–Zn–Te region;
- reduce or remove Bi based on the SHAP trend;
- add phase-stability and formation-energy filters before final recommendation;
- combine zT, mechanical stability, toxicity and raw-material cost into a multi-objective optimization function;
- validate the top three compositions experimentally.

13 References and Submitted Materials

1. Jang et al., "Active Learning-Guided Accelerated Discovery of Ultra-Efficient High-Entropy Thermoelectrics," *Advanced Materials*, 2026, e15054.
2. ESTM thermoelectric database, local file: `estm.xlsx`; public source referenced in the presentation: `https://github.com/KRICT-DATA/SIMD`.
3. Project presentation: `new presentation file.pdf`.
4. Project code: `new.py` and Google Colab notebook `https://colab.research.google.com/drive/1mQFy-uFSYemwAepVUa0JbUd-DotyVMTa`.

A Appendix: Code Workflow Summary

The code follows a reproducible sequence:

1. Load ESTM data and clean rows with missing thermoelectric properties.
2. Convert chemical formulas into numerical composition features using matminer and pymatgen.
3. Train Random Forest models for thermoelectric property prediction.
4. Build a candidate design space of high-entropy chalcogenide compositions.
5. Evaluate candidates through a surrogate oracle.

6. Select initial samples and run batch active learning using Gaussian Process Regression.
7. Rank candidates using an average performance target.
8. Generate SHAP and t-SNE visualizations for interpretation.

This structure is suitable for further extension because new experimental measurements can be appended to the dataset and the active learning loop can be rerun with improved confidence.